

No. of Functional Features Included in the Solution

Date	9 July 2026
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Functional Features Overview

This document lists all the functional features implemented in the Credit Card Approval Prediction System. Each feature contributes to automating the credit card approval process using machine learning and a Flask-based web application.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	User Input Form	Accepts applicant information through the web interface.	Frontend (HTML/CSS/Bootstrap)	Done	High
2	Data Preprocessing	Cleans data, handles missing values, removes duplicates, and encodes categorical features.	Data Processing	Done	High
3	Machine Learning Model	Trains Logistic Regression, Decision Tree, Random Forest, and XGBoost models.	ML Module	Done	High
4	Model Evaluation	Compares models using Accuracy, Confusion Matrix, Precision, Recall, and F1-Score.	Evaluation Module	Done	High
5	Real-Time Prediction	Predicts whether a credit card application will be approved or rejected.	Prediction Engine	Done	High
6	Flask Web Application	Integrates frontend with backend to provide real-time prediction.	Flask Backend	Done	High
7	Model Storage	Saves and loads the trained model using Joblib.	Model Management	Done	Medium
8	Cloud Deployment Support	Supports deployment using IBM Watson Machine Learning / Render.	Deployment	In Progress	Medium

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	7

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Applicant Form, Prediction Result Page
2	Backend / Logic	3	Data Preprocessing, Model Prediction, Flask Routing
3	Database / Storage	2	CSV Dataset, Trained Model (model.pkl)
4	API / Integration	1	Flask API, IBM Watson ML Integration
5	Security / Authentication	1 (Future)	User Login & Authentication